

Computational Physics and Data Science Exercise 4: Pleiades Cluster H-R Diagram

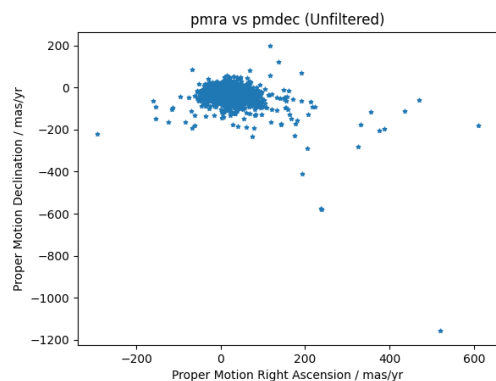
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INTRODUCTION

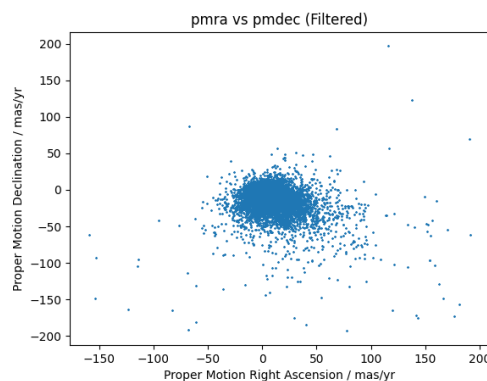
This exercise is based on data acquired from the Gaia space telescope mission run by the ESA. A portion of the approximately 2×10^9 stars surveyed are used in the following exercise [1]. The target region is M45/Pleiades, a region with data on 113,629 stars. The three principal goals of this experiment are to isolate an 'interesting' cluster for analysis, perform principal component analysis, and then recover a Hertzsprung-Russell diagram from the remaining stars.

I. READING AND FILTERING DATA

The raw dataset of 113,629 stars is described by 12 different features, namely Right Ascension, Declination, Galactic Longitude Galactic Latitude, Parallax, Proper Motion in Right Ascension, Proper Motion in Declination, G-band Magnitude, Blue Photometric Band Magnitude, Red Photometric Band Magnitude, Absolute Uncertainty in Parallax, and Renormalised Unit Weight Error. We begin this section by filtering the data to remove extreme/invalid data points. Parallax is by definition positive, so values below zero must be noise with no physical significance; these are removed. We also remove absolute values of `pmra` and `pmdec` of greater than 200 mas yr^{-1} as these are physically implausible motions. Finally we remove values with a greater than 20% absolute uncertainty in parallax, as this will broaden our clusters in the following analysis stages. Applying `dropna()` is required for the matrix operations used in PCA, so this is also done.



(a) A graph of the unfiltered pmra vs pmdec data. The central core dominates, but a long tail is visible reaching several hundreds of mas/yr in both axes, obscuring the interesting structure.



(b) A graph of the filtered pmra vs pmdec data. The central cluster is much more visible. All these bodies share similar proper motions and thus form a cluster in space.

FIG. 1: We can clearly see that the extreme values have been removed through the filtering process. This leaves a large diffuse background with a central clump. This central clump is the Pleiades.

II. STANDARDISING

Each feature of the data spans over a vast range of magnitudes and units. Performing PCA on this data would cause values with large variance to dominate the principal components, masking the structure

of interest. Using `StandardScaler` from `sklearn` transforms each feature as follows.

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

where z is the transformed feature, x is the untransformed feature, μ is the mean of the column, and σ its standard deviation. This yields a data set with zero mean and a variance of one for each variable, which allows each feature to be equally considered in PCA.

III. CLUSTERING

Clustering is performed on the subspace of proper motion (`pmra` and `pmdec`) of the scaled data. This is done since the Pleiades is a single structure with motion that should be distinguishable from the background in this space. K-means groups similar points by finding the centre of each group and then moving points to their nearest centre until the groups are constant. This method assumes approximately spherical clusters of approximately equal size. We can see in Figure 1b that the entire distribution is not of this form - it is irregular and elongated. This will cause K-means to force linear boundaries through data in which this makes no physical sense. With $n = 2$ we see a split along no meaningful axis (figure 2), separating the data. This is due to the background data overpresiding over the calculation. We can

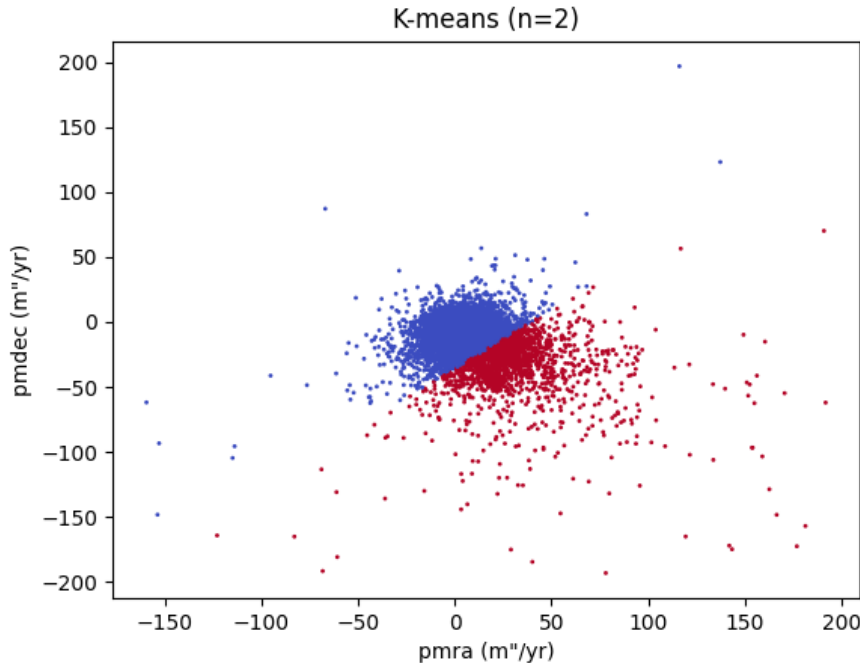


FIG. 2: The scaled data split using K-means with $n = 2$. We see here a sharp line splitting the data in no meaningful way.

increase the value of n to $n = 10$ to improve background consolidation, but we still see a splitting of our cluster of interest in a way that does not recover one component (figure 3).

The Gaussian Mixture Model (GMM) suits this task much better. In GMM, each component is treated as a Gaussian with its own mean, C matrix, and mixing weight. This can generate elliptical divisions in the data of any orientation and size, therefore being much more flexible to true physical behaviours that K-means can be. The background distribution is both non-spherical and multi-modal (a Gaussian with multiple peaks). With sufficient components (in this case, $n = 10$), the model has enough room to describe the background, separating the defined cluster locus. This method has limits, as with $n = 2$ the model cannot isolate the background, so compromises the integrity of the cluster of interest. Increasing n , therefore, allows for a more cohesive background model, freeing up a single component to account for the physical behaviour of the Pleiades cluster. The GMM with $n = 10$ isolates 1034 stars, just 0.91% of the original unfiltered dataset. This is a plausible for an open star cluster viewed against a dense background field. The Pleiades is a gravitationally bound cluster of stars formed from the same molecular cloud.

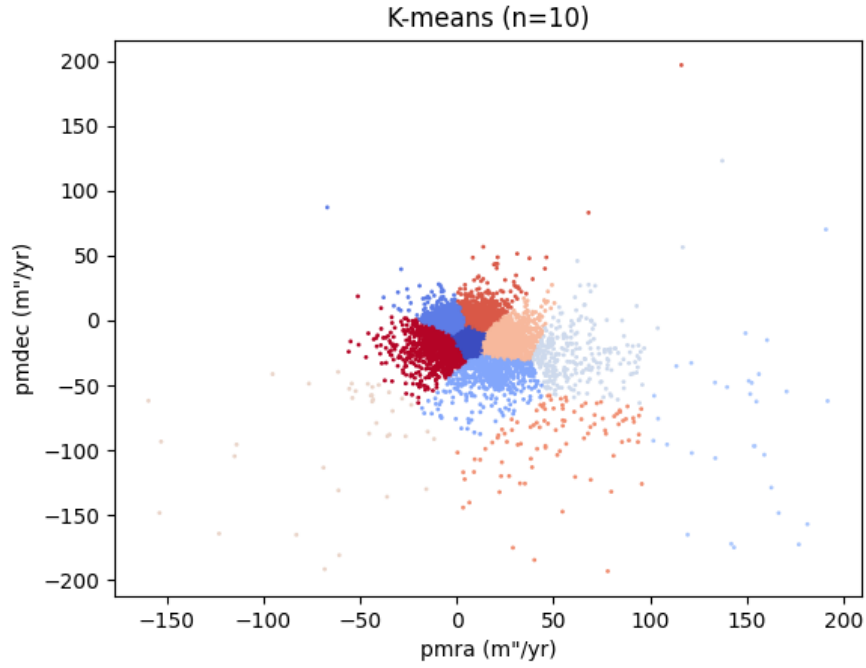


FIG. 3: The scaled data split using K-means with $n = 10$. There is a much greater discernment in background vs cluster, but the cluster itself is split into physically unmeaningful sections.

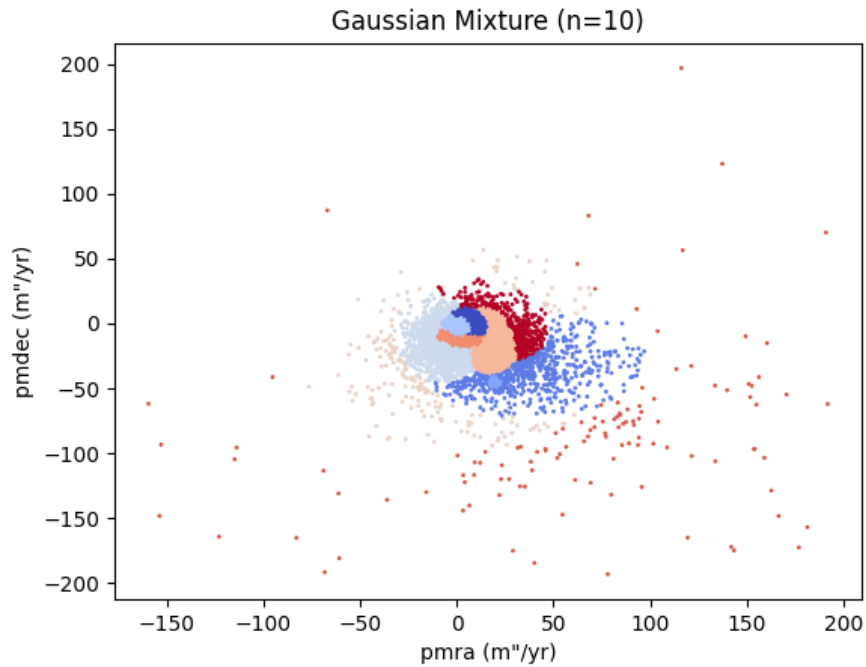


FIG. 4: The scaled data split using GMM with $n = 10$. We can see the elliptical boundaries that have isolated well the nucleus of the Pleiades cluster (at $pmra, pmdec \approx 25, -40$), as well as properly isolating the background.

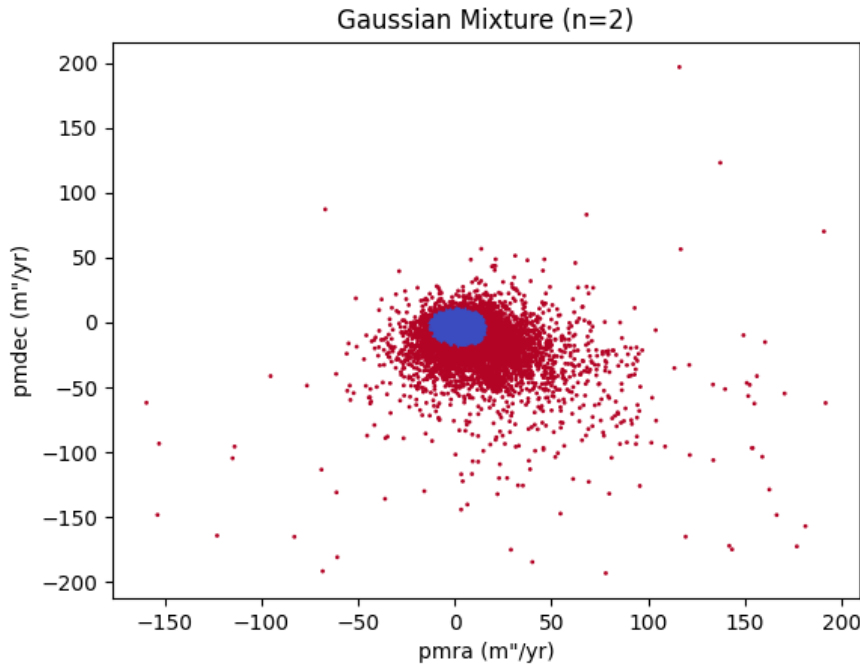


FIG. 5: The scaled data split using GMM with $n = 2$. The model fails to discern the background from the cluster of interest.

They, therefore, have the same velocity through the Milky Way. The physical interpretation of this from earth is a common proper motion, whereby all of these stars appear at roughly the same pmra-pmdec coordinate. This is what GMM at $n = 10$ has managed to isolate.

IV. PRINCIPAL COMPONENT ANALYSIS

PCA was applied to solely the cluster of interest (1034 stars) that was identified in section III.

As there are 12 features, a 12×12 covariance matrix $\mathbf{C}$ is computed from the cluster of interest. Eigendecomposition $\mathbf{C} = \mathbf{V}\mathbf{D}\mathbf{V}^T$ is performed, where the columns of $\mathbf{V}$ are the principal directions, and the diagonal matrix $\mathbf{D}$'s elements are the eigenvalues, which are proportional to variance.

principal Component (PC) 1 explains 27.78% of the variance and PC2 explains 18.97%, meaning that the first two principal components account for 46.75% of the variance. The gap between PC1 and PC2 is not very dramatic, an expected outcome of the data standardisation; no single feature dominates the dataset numerically. The properties of the stars are therefore genuinely multi-dimensional, as the other 10 components count for over 50% of the variance - this cannot be neglected. Plotting PC1 against PC2 (as in figure 6) we see zero trend or correlation, the stars simply present a uniform cloud. This is due to the fact that each star in the cluster has already been identified as being in the Pleiades.

V. PROJECTION AND FACTOR ANALYSIS

The loadings of the dataset can be found via

$$f_x^i = v_x^i \sqrt{\lambda_x} \quad (2)$$

where f_x^i are the loadings for a component x for a feature i , where λ_x is the explained variance.

Table I shows the top three features for each principal component. We can see that PC1 is dominated by optical components, with red photometric band, G-band, and blue photometric band magnitudes being the most important. These features map to luminosity and surface temperature - the axes of a Hertzsprung-Russell diagram. PC2 on the other hand is dominated by spacial components, with right ascension and galactic longitude being the most dominant. This shows that PC2 encodes the spatial properties and distance structure of the Pleiades.

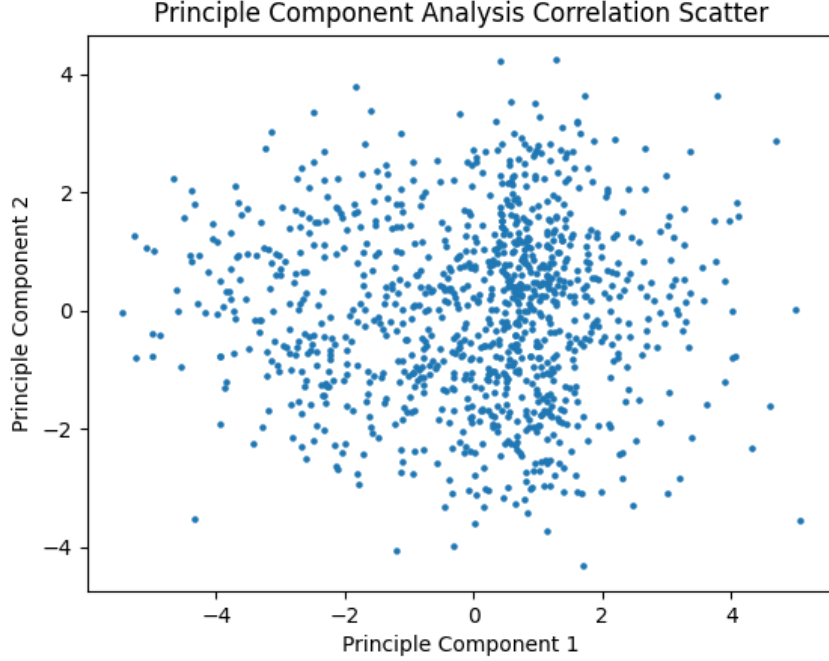


FIG. 6: A plot of PC1 against PC2. There is no trend between the two, with the stars forming a uniform cloud.

Top 3	Feature	PC1 Loading	PC2 Loading
PC1	rp	0.990183	-0.008592
	g	0.988606	-0.013115
	bp	0.981852	-0.015172
PC2	ra	-0.035328	0.906994
	l	0.010671	0.849804
	pmra	-0.064042	-0.659143

TABLE I: Top 3 Features by Loadings for Principal Components 1 and 2

We can use the G-band magnitude as apparent magnitude, and colour-index (blue photometric band – red photometric band) as a substitute for surface temperature (these values are correlated). Plotting apparent magnitude on the y-axis (inverted, high to low) and colour index on the x (figure 7), we can see a graph that is very reminiscent of a Hertzsprung-Russell diagram. The main line visible is very firmly main sequence. The Pleiades is a young star cluster (~ 115 Myr [2]), which explains the main sequence dominance - there has simply not been enough time for further stellar evolution. We can see some of the stars on the brighter end decreasing in colour index - these are likely more massive stars beginning to reach the end of their lives. This graph shape proves that the GMM pipeline was effective in isolating the Pleiades cluster from the background.

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- [1] Woodgate, C. m45-clean.csv: A cleaned dataset for pleiades (m45) stellar photometry. <https://github.com/ChrisWoodgate/demo-files/blob/main/csv/m45-clean.csv> (2022). GitHub Repository: demo-files.
- [2] Basri, G., Marcy, G. W. & Graham, J. R. Lithium in brown dwarf candidates: The mass and age of the faintest pleiades stars. *The Astrophysical Journal* **458**, 600–609 (1996). Bibcode: 1996ApJ...458..600B.

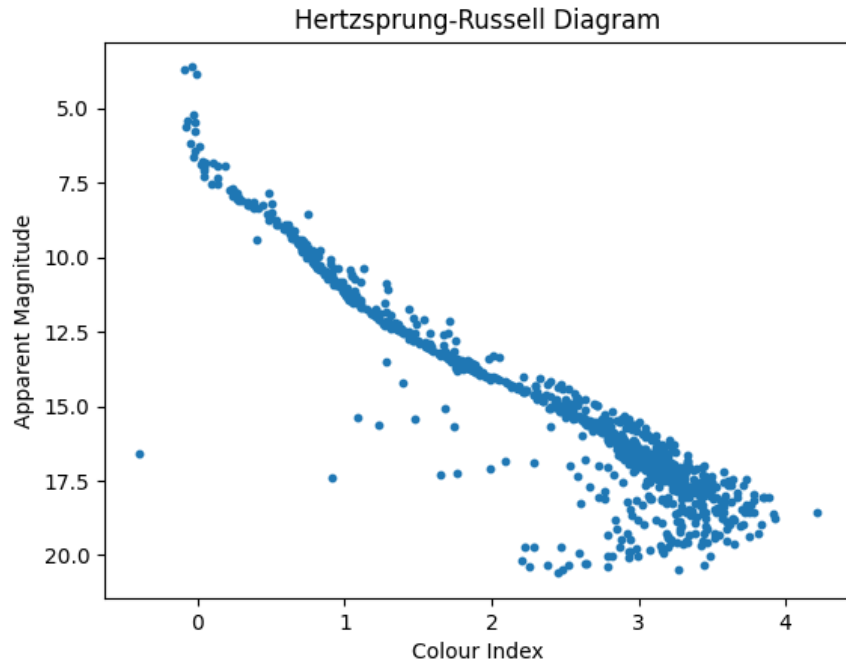


FIG. 7: A plot of Apparent Magnitude (g) vs colour index ($bp-rp$). A clear line on the diagonal is present. This line is the line of main sequence stars seen on a Hertzsprung-Russell diagram.